

WEB DESIGN AND PROGRAMMING PROJECT

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Project Description

BookStore is a modern full-stack e-commerce web application designed for selling books online. The project implements a complete three-tier architecture with a React-based frontend, Spring Boot REST API backend, and PostgreSQL database.

The application provides users with an intuitive interface to browse books, view detailed information, search for specific titles, and manage a shopping cart. The frontend is built using React 18.2.0 with Material UI components, ensuring a responsive and modern user experience across all devices. The backend utilizes Spring Boot 3.2.0 with Spring Data JPA for efficient data persistence and RESTful API endpoints.

Important Features

- **Modern Frontend:** A fully responsive UI built with **React** and **Material UI 5.14**, featuring custom themes and **React Router** for seamless navigation.
- **Persistent Shopping Cart:** Managed via **Context API** with **LocalStorage** integration, ensuring items persist after refresh. Includes stock validation and real-time price calculation.
- **RESTful Spring Boot Backend:** A robust API supporting **CRUD operations**, utilizing **DTOs** for clean data transfer and structured error handling.
- **Advanced SQL Search:** High-performance search functionality across titles, authors, and descriptions, delivering real-time results in a grid layout.
- **Detailed Product Pages:** Dedicated pages for each book featuring full metadata (ISBN, stock status, and cover images) with direct "add to cart" integration.

Technical Installation

- **Database:** Import the SQL file located in the `/database` folder to your SQL server.
- **Backend:** Open the `/backend` folder in **IntelliJ IDE** and run the project using **Maven**.
- **Frontend:** Navigate to the `/frontend` folder in your terminal and run the following commands:

```
npm install (to install dependencies)
```

```
npm start
```